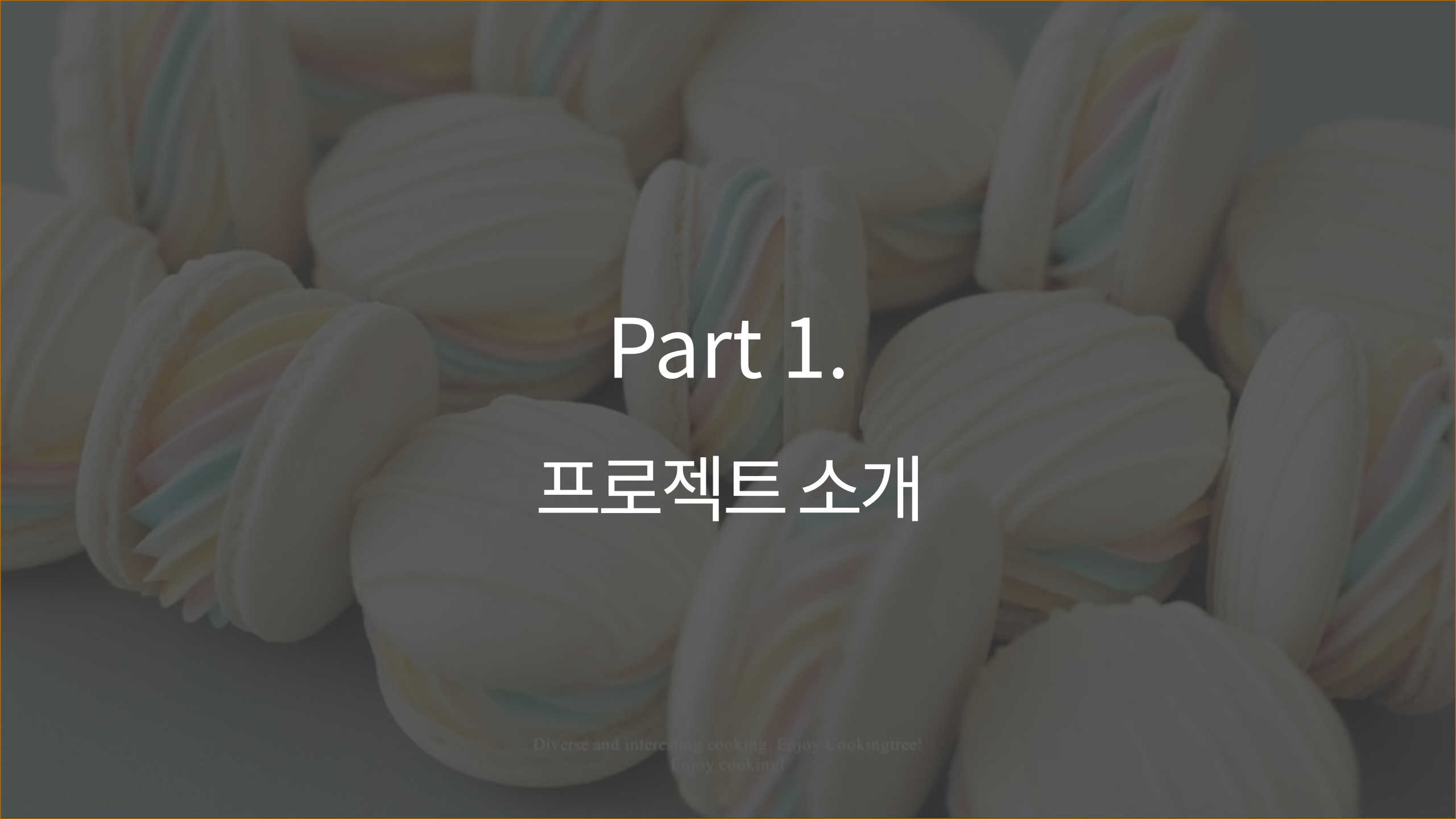


마카롱맛 프로젝트

최승열

Feat. Kubernetes로 개발서버 간단히 짚어내기



The background of the slide is a close-up photograph of numerous macarons. The macarons are primarily white with delicate, multi-colored swirls of pastel colors (pink, yellow, blue, and green) visible within their shells. They are arranged in a dense, overlapping pattern, creating a soft and appetizing texture. The lighting is soft, highlighting the smooth surface of the macarons.

Part 1.

프로젝트 소개

Diverse and interesting cooking. Enjoy Cookingtree!
Enjoy cooking!

구현한 기능



서버 생성/삭제의
간편화



동적 서브도메인 생성
(Ingress 활용)



서버 관리용 대시보드

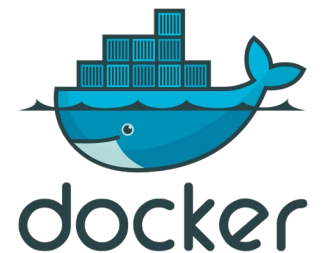
기술스택

Container Orchestration

- Kubernetes, containerd, Docker

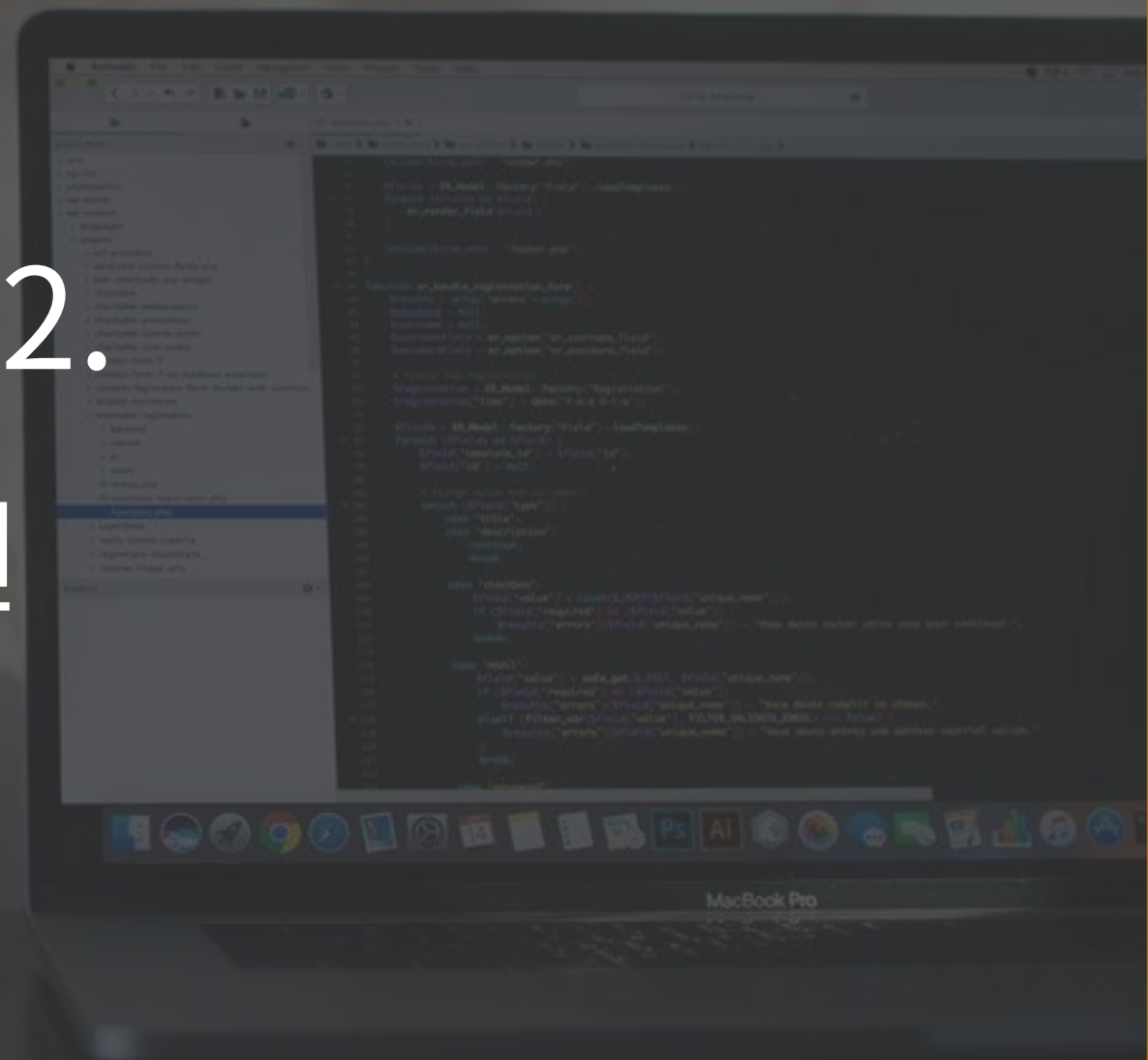
서버 관리용 대시보드

- Python, FastAPI, Pykube, MySQL



Part 2.

구현

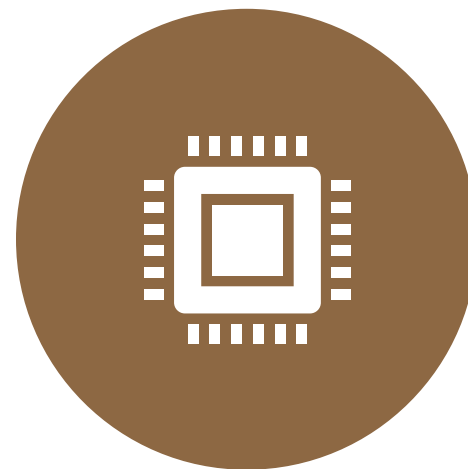


STEP 1. 서버 환경세팅



작업서버

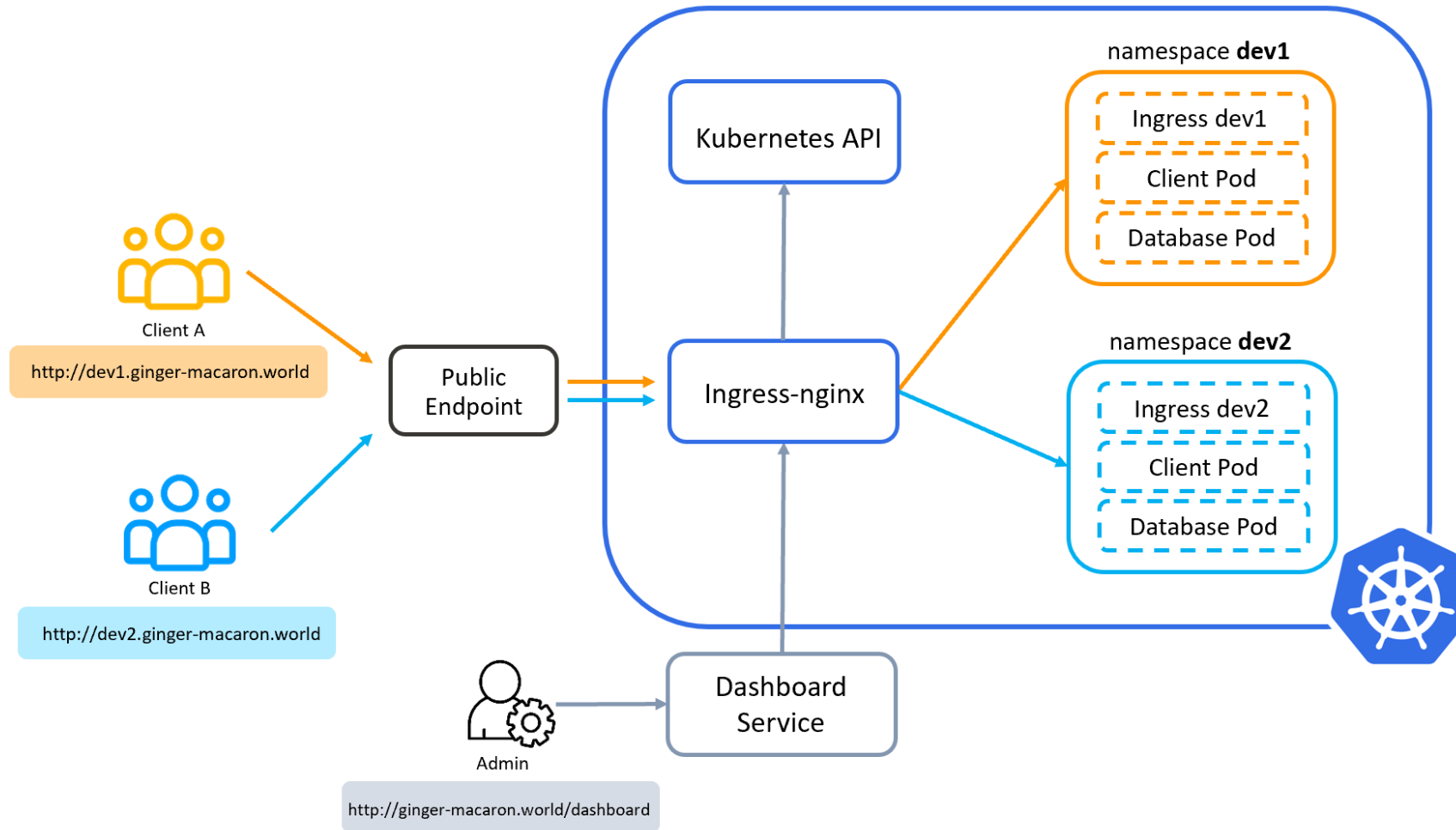
개인 홈 서버 (물리)
가상화 하이퍼바이저 (Proxmox)



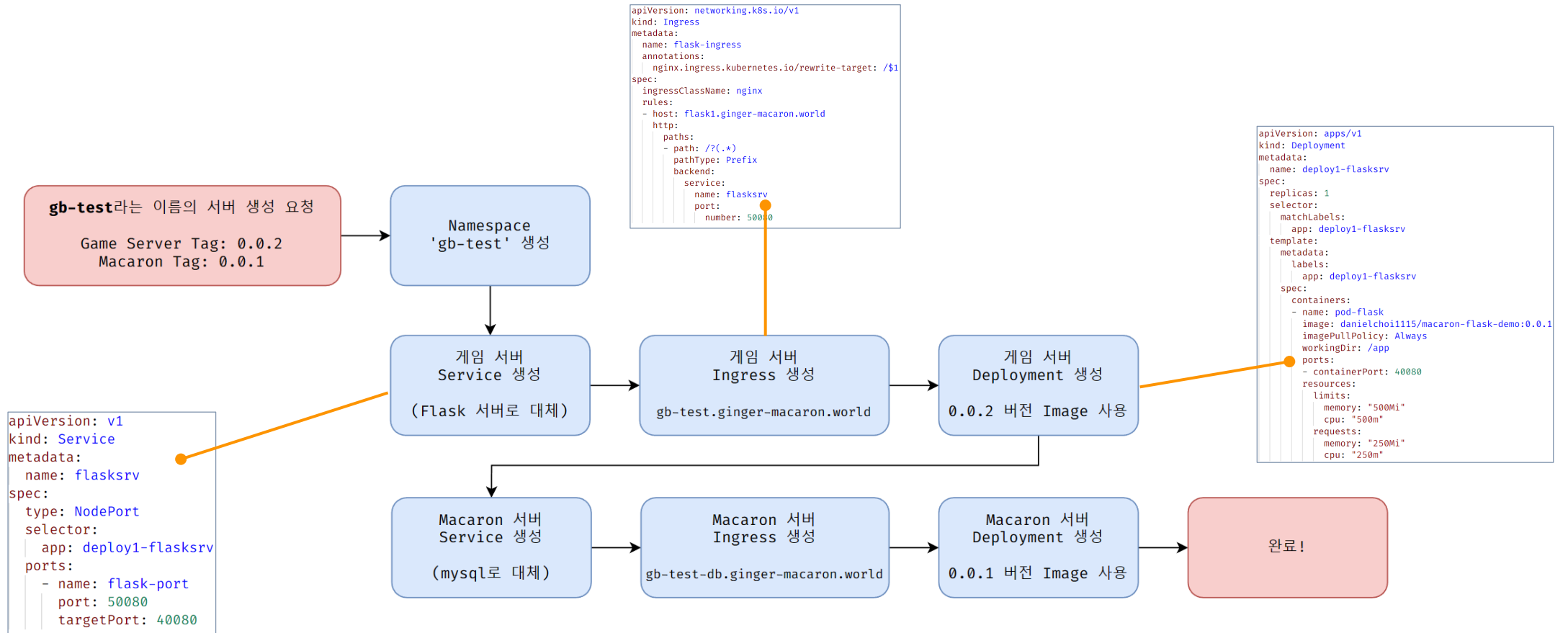
Kubernetes 환경

Ubuntu 22.04 VM 기반
Master 노드 1대 + Worker 노드 2대

STEP 2. Kubernetes 구조 작성



STEP 3. Kubectl을 사용한 서버 생성 흐름도



STEP 4. 대시보드 구현

Macaron Dashboard



개발자 Y

서버 생성



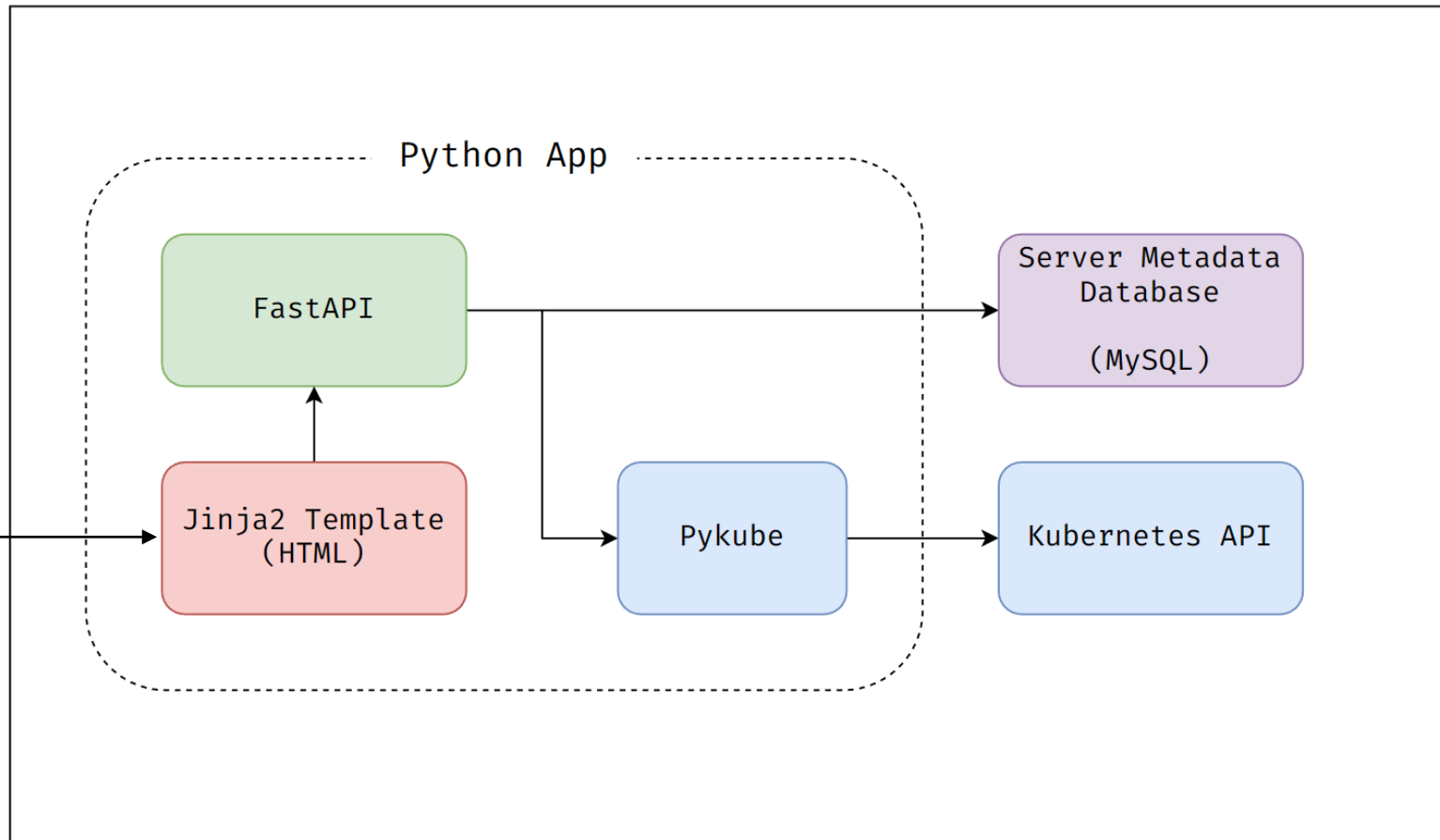
개발자 H

서버 삭제



개발자 N

서버 정보 확인



Part 3.

프로젝트 데모

서버를 띄우는 Dashboard

ginger-macaron.world/dashboard

ADMIN

Dashboard

Notifications

Support

Servers

+ Launch New Server

신규 서버 생성 버튼

My server

#	SERVER NAME	AGE	DESCRIPTION	DEPLOYER	GAME SERVER TAG	MACARON TAG	STATUS	ACTION
1	gb-test2	Just now	두번째 테스트 서버	macaron	0.0.2	0.0.1	Running	Delete

Other dudes' server

#	SERVER NAME	AGE	DESCRIPTION	DEPLOYER	GAME SERVER TAG	MACARON TAG	STATUS	ACTION
1	gb-test	7 days	Other test server	ginger	0.0.1	0.0.1	Running	-

서버 생성화면

Create New Server

×

Server Information

Server Name

gb- macaron1

You can use this server name!

Server Description

Macaron의 서버 1

Server Deployer

macaron

Game Server Tag

Client Version 0.0.2

Macaron Tag

Database Version 0.0.1

Create

서버 정보 입력

생성된 서버

Servers

+ Launch New Server

☰ My server

#	SERVER NAME	AGE	DESCRIPTION	DEPLOYER	GAME SERVER TAG	MACARON TAG	STATUS	ACTION
1	gb-macaron1	Just now	Macaron의 서버 1	macaron	0.0.2 ↗	0.0.1 ↗	Running	Delete

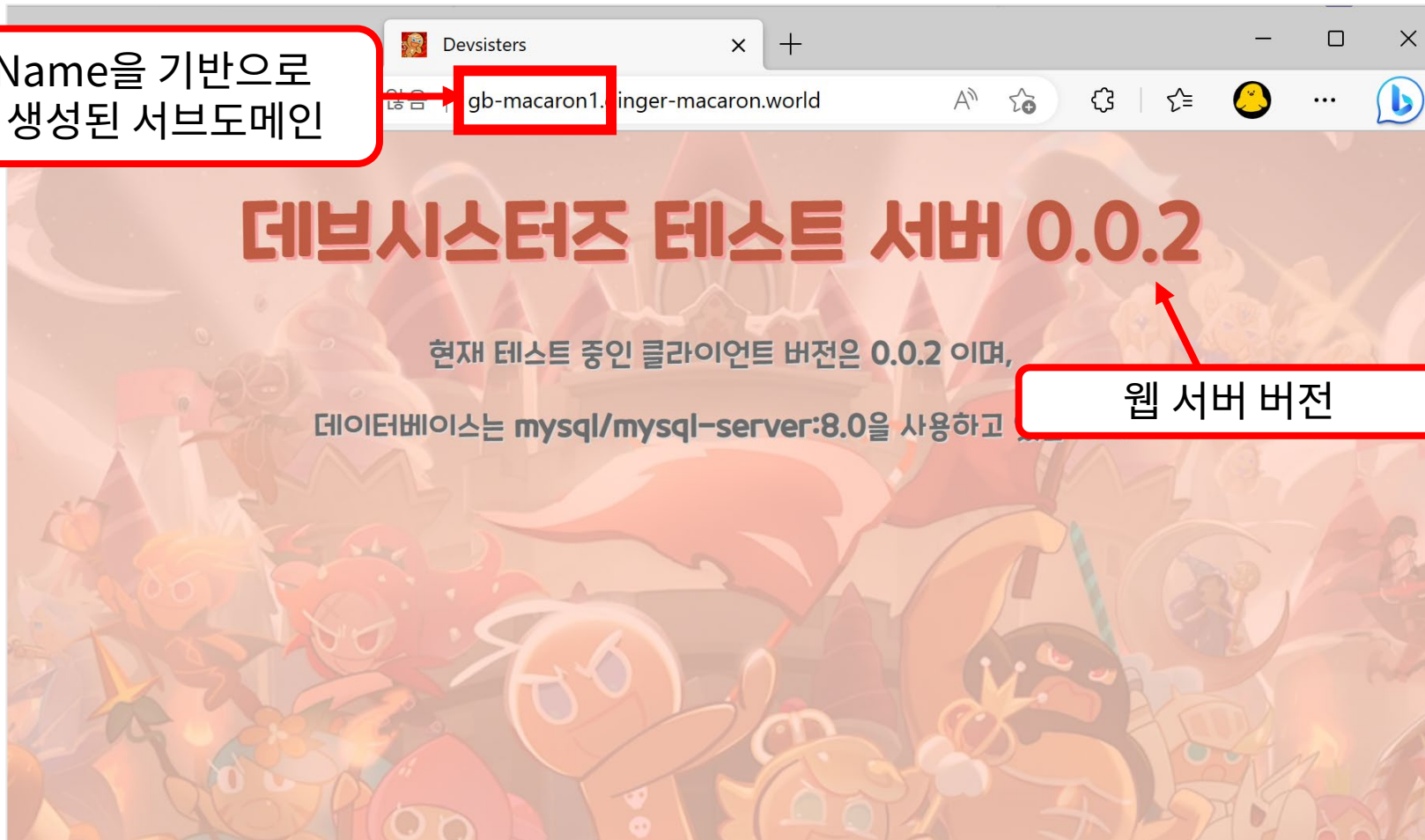
웹 서비스 링크

서버 메타정보

서버상태

웹 서비스 접근

Server Name을 기반으로
동적으로 생성된 서브도메인



웹 서버 버전

서버 삭제

☰ My server

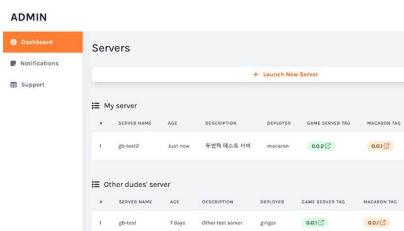
#	SERVER NAME	AGE	DESCRIPTION	DEPLOYER	GAME SERVER TAG	MACARON TAG	STATUS	ACTION
1	gb-macaron1	Just now	Macaron의 서버 1	macaron	0.0.2 ↗	0.0.1 ↗	Terminating	Delete

서버 삭제 버튼

☰ My server

#	SERVER NAME	AGE	DESCRIPTION	DEPLOYER	GAME SERVER TAG	MACARON TAG	STATUS	ACTION
1	gb-macaron1	Just now	Macaron의 서버 1	macaron	0.0.2 ↗	0.0.1 ↗	Deleted	Delete

대시보드 및 데모영상 링크



대시보드 링크

<http://ginger-macaron.world/dashboard>



데모영상 유튜브 링크

<https://www.youtube.com/watch?v=XeAOgATG0aU>

감사합니다!

